

Ayon Aryan

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SUMMARY

Computer Science undergraduate at RV University (AI/ML major, Fintech minor) with two completed industry internships shipping production AI and mobile systems. Built a legal-domain Retrieval-Augmented Generation service with prompt-injection defense at Broadrange AI, and a 100+ install Play Store Android app at Techpuram Technology. Strong foundation in Data Structures, Object-Oriented Programming, and Database Management Systems; comfortable owning features end-to-end in Python, Java, and Kotlin.

EDUCATION

RV University, Bangalore — B.Tech (Hons.) Computer Science Expected May 2027

Major: Artificial Intelligence & Machine Learning · Minor: Fintech · CGPA: **7.8 / 10** · *Best Project Award (Structural Innovation)* · IEEE CCEM 2024 paper presenter

Gyansthal High School, Patna — Class XII (CBSE), 2023 · **Delhi Public School, Patna — Class X (CBSE), 2022**

TECHNICAL SKILLS

- **Languages:** Python, Java, Kotlin, C++, C, SQL
- **AI / ML:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, SHAP, Real-ESRGAN, BERT, CNNs
- **LLM / RAG:** Retrieval-Augmented Generation, Graph-RAG, BERT intent classification, prompt-injection defense (PromptGuard V2), Whisper, Ollama, Groq, SambaNova
- **Backend:** Spring Boot, Flask, REST APIs, Microservices, MVC, Maven, Gradle, JUnit, Postman
- **Databases:** PostgreSQL (pgvector), MySQL, SQLite, MongoDB, Cassandra, Redis, ChromaDB, FAISS
- **Mobile / Cloud / DevOps:** Jetpack Compose, CameraX, Google Maps SDK, Play Console; Docker, Kubernetes fundamentals, AWS CLI, Linux, Git, GitHub Actions
- **Concepts:** DSA, OOP, DBMS, Operating Systems, Computer Networks, System Design fundamentals

EXPERIENCE

Broadrange AI — Artificial Intelligence Intern Chicago, IL (Remote) · Jun 2025 – Aug 2025

- Architected a legal-domain RAG pipeline (Flask + ChromaDB + sentence-transformer embeddings + SambaNova Llama-4) over a curated corpus, eliminating off-domain hallucinations on internal eval prompts.
- Integrated **PromptGuard V2** at request ingress to block prompt-injection and jailbreak attempts before any LLM call, hardening the public API surface.
- Designed chunking-based document ingestion with metadata-aware retrieval and domain guardrails, improving grounded-answer top-k recall and reducing irrelevant retrievals.
- Delivered the pipeline end-to-end: ingestion → embeddings → retrieval → generation → validation → REST API.

Techpuram Technology Pvt. Ltd. — Android Native App Developer Intern Madurai, Tamil Nadu · Feb 2025 – Aug 2025

- Shipped **Geo GPS Camera**, a tamper-evident location-tagged photography app (Kotlin, Jetpack Compose, CameraX, Google Maps Compose SDK); reached **100+ Play Store installs** with zero crash regressions in the rollout window.
- Implemented server-side validation to detect spoofed images and falsified GPS metadata, hardening the upload-to-verification flow against fake-location attacks.
- Owned the full Android stack: Compose UI, runtime permissions, camera capture, Maps SDK, REST integration, Gradle release builds, and Play Console rollout.

SELECTED PROJECTS (GITHUB.COM/AYON-ARYAN)

- **Image Resolution Enhancer (Face Super-Resolution)** — Fine-tuned **Real-ESRGAN** on a paired HR-LR face dataset using transfer learning and custom loss balancing; improved **PSNR from 29.15 dB → 33.95 dB** with SSIM 0.8985. *PyTorch, OpenCV*
- **Local NL-to-SQL Database Manager** — Offline natural-language-to-SQL gateway across **8 database engines** (SQLite, MySQL, PostgreSQL, MSSQL, Oracle, MongoDB, Cassandra, Redis); dual-LLM routing (Groq + offline Ollama), schema-aware prompting, human-in-the-loop write review, snapshot rollback, RBAC, Fernet-encrypted credentials. *Python, SQLAlchemy*
- **LLM-Service-MCP (Autonomous Agent Platform)** — Microservice AI agent with a BERT intent classifier routing prompts dynamically across RAG, web-search, and chat tools. *Java, Spring Boot, REST*
- **Credit-Risk-XAI** — Explainable credit underwriting with XGBoost + LightGBM, SHAP global/local explanations, FairLearn fairness auditing, FCRA-style adverse-action notice generation. *Python, Streamlit*
- **Graph-RAG** — Hybrid retrieval combining NetworkX knowledge-graph traversal, FAISS vector search, and Groq Llama inference with an interactive D3.js graph visualization. *Python, JavaScript*

CERTIFICATIONS & ACHIEVEMENTS

- **IEEE CCEM 2024** — Paper presenter, *GlucoseSense*: non-invasive saliva glucose biosensor · **Best Project Award (Structural Innovation)**, RV University
- **AWS Academy** — Machine Learning Foundations · CLI Basics · Advanced DevOps Testing Practices
- **Google Cloud** — Generative AI Fundamentals · Introduction to Large Language Models · Introduction to Responsible AI
- **IBM SkillsBuild** — Big Data 101 · Hadoop 101 · Spark Fundamentals · Containers, Kubernetes & OpenShift
- **NPTEL** Programming in Modern C++ · **Scaler DSA** · **Coursera** Machine Learning · **HackerRank** Java (Basic) · **Cisco** Packet Tracer · **Microsoft & LinkedIn** Career Essentials in Business Analysis
- **LeetCode** 116 solved (39E · 64M · 13H) · **Kaggle** Python · **CodeChef** rated · **Y20 Summit** (G20 India 2023) youth delegate